REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 10 Advanced | Module 16 Lesson 2 | Curved Mirrors

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: f = R/2; 1/f = 1/do + 1/di; m = hi/ho = −di/do. Concave f positive; convex f negative; real di positive.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 9–18.

P01. A concave mirror can form a real image when the object is:

A. At every possible position B. Always between mirror and focus
C. Beyond its focal point D. Behind the mirror only

Answer

C. Beyond its focal point.

P02. A convex mirror forms images of real objects that are:

A. Real inverted and enlarged B. Real upright and diminished
C. Virtual inverted and enlarged D. Virtual upright and diminished

Answer

D. Virtual upright and diminished.

P03. For a spherical concave mirror, focal length is approximately:

A. Half the radius of curvature B. Twice the radius of curvature
C. Equal to object height D. Independent of curvature

Answer

A. Half the radius of curvature.

P04. A negative magnification means the image is:

A. Always virtual B. Inverted
C. Always larger D. At infinity

Answer

B. Inverted.

P05. A real object at the centre of curvature of a concave mirror has an image that is:

A. Virtual upright and enlarged B. Real upright and diminished
C. Real inverted and equal in size D. Virtual inverted and equal in size

Answer

C. Real inverted and equal in size.

P06. A spherical concave mirror has radius of curvature 24 cm. Calculate its focal length.

A. 12 cm B. 24 cm
C. 6 cm D. -12 cm

Answer

A. 12 cm. f = R/2 = 24/2 = 12 cm.

P07. A concave mirror has focal length +10 cm. An object is 30 cm in front. Calculate its signed image distance using 1/f = 1/do + 1/di.

A. 30 cm B. 15 cm
C. 7.5 cm D. -15 cm

Answer

B. 15 cm. 1/di = 1/10 − 1/30; di = 15 cm. Positive di means a real image in front.

P08. A convex mirror has f = -10 cm and object distance do = +20 cm. Find the signed image distance.

A. -13.33 cm B. -3.333 cm
C. -6.667 cm D. 6.667 cm

Answer

C. -6.667 cm. 1/di = 1/f − 1/do = 1/(-10) − 1/20; di = -6.667 cm. The image is virtual behind the mirror.

P09. A mirror forms a real image 12 cm in front when the object is 24 cm in front. Calculate its signed magnification.

A. -1 B. -0.25
C. 0.5 D. -0.5

Answer

D. -0.5. m = −di/do = −12/24 = -0.5. Negative means inverted; |m| < 1 means diminished.

P10. A mirror has magnification −0.50 for an object 6 cm tall. Find the signed image height.

A. -3 cm B. -6 cm
C. -1.5 cm D. 3 cm

Answer

A. -3 cm. hi = mho = −0.50 × 6 = -3 cm. The image is inverted.

P11. Define principal axis of a spherical mirror.

Answer

The line passing through the mirror's vertex and centre of curvature.

P12. Define focal point of a concave mirror.

Answer

The point where paraxial rays initially parallel to the principal axis converge after reflection.

P13. Define focal length.

Answer

The distance along the principal axis from the mirror vertex to its focal point, with the sign determined by convention.

P14. Why are convex mirrors useful for vehicle observation?

Answer

They form upright diminished images and provide a wider field of view than a comparable plane mirror.

P15. What does positive signed image distance mean under the stated mirror convention?

Answer

A real image forms in front of the mirror, where the reflected rays actually converge.

P16. A spherical concave mirror has radius of curvature 36 cm. Calculate its focal length.

Answer

f = R/2 = 36/2 = 18 cm.

P17. A spherical concave mirror has radius of curvature 48 cm. Calculate its focal length.

Answer

f = R/2 = 48/2 = 24 cm.

P18. A concave mirror has focal length +15 cm. An object is 45 cm in front. Calculate its signed image distance using 1/f = 1/do + 1/di.

Answer

1/di = 1/15 − 1/45; di = 22.5 cm. Positive di means a real image in front.

P19. A concave mirror has focal length +20 cm. An object is 60 cm in front. Calculate its signed image distance using 1/f = 1/do + 1/di.

Answer

1/di = 1/20 − 1/60; di = 30 cm. Positive di means a real image in front.

P20. A convex mirror has f = -15 cm and object distance do = +30 cm. Find the signed image distance.

Answer

1/di = 1/f − 1/do = 1/(-15) − 1/30; di = -10 cm. The image is virtual behind the mirror.

P21. A convex mirror has f = -20 cm and object distance do = +40 cm. Find the signed image distance.

Answer

1/di = 1/f − 1/do = 1/(-20) − 1/40; di = -13.33 cm. The image is virtual behind the mirror.

P22. A mirror forms a real image 18 cm in front when the object is 36 cm in front. Calculate its signed magnification.

Answer

m = −di/do = −18/36 = -0.5. Negative means inverted; |m| < 1 means diminished.

P23. A mirror forms a real image 24 cm in front when the object is 48 cm in front. Calculate its signed magnification.

Answer

m = −di/do = −24/48 = -0.5. Negative means inverted; |m| < 1 means diminished.

P24. A mirror has magnification −0.50 for an object 9 cm tall. Find the signed image height.

Answer

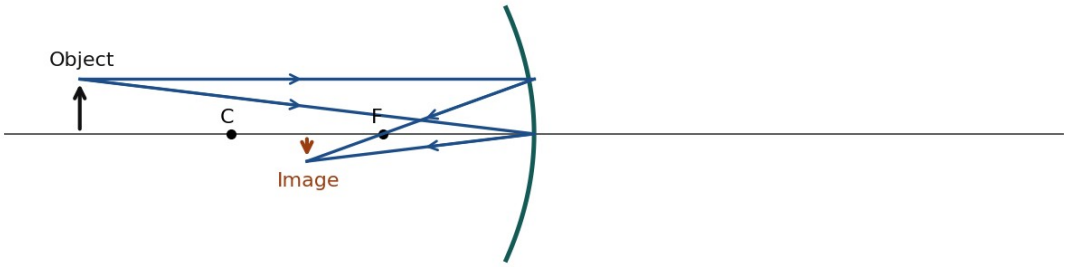
hi = mho = −0.50 × 9 = -4.5 cm. The image is inverted.

P25. A mirror has magnification −0.50 for an object 12 cm tall. Find the signed image height.

Answer

hi = mho = −0.50 × 12 = -6 cm. The image is inverted.

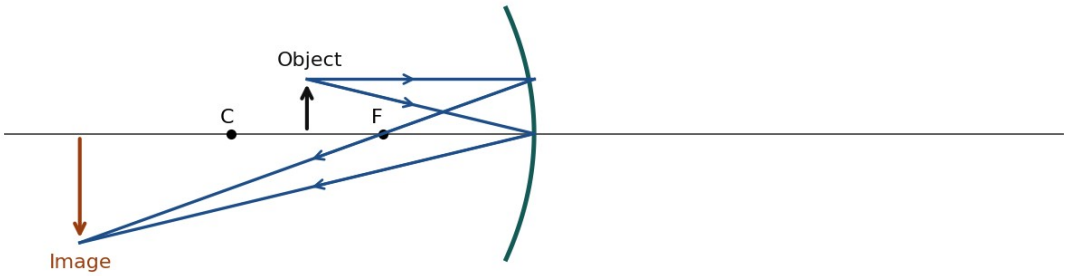
P26. Draw a ray diagram for an object beyond C of a concave mirror. State the image properties.



Answer

A parallel incident ray reflects through F; a ray through C returns along itself. The image lies between F and C and is real, inverted and diminished.

P27. Draw a ray diagram for an object between F and C of a concave mirror.



Answer

The reflected principal rays converge beyond C. The image is real, inverted and enlarged.

P28. A concave mirror is used as a magnifying shaving mirror. Where should the face be relative to F? Explain.

Answer

Place the face between the mirror and its focal point. Reflected rays diverge and their backward extensions form a virtual, upright, enlarged image.

P29. A student claims that any image larger than the object must be virtual. Evaluate the claim.

Answer

False. A concave mirror with its object between F and C forms an enlarged real image. Image size alone does not determine whether an image is real.

P30. Why should a ray diagram be checked against the sign of calculated magnification?

Answer

Negative magnification predicts an inverted image; positive magnification predicts an upright image. A disagreement can reveal an error in the diagram, signs or algebra.